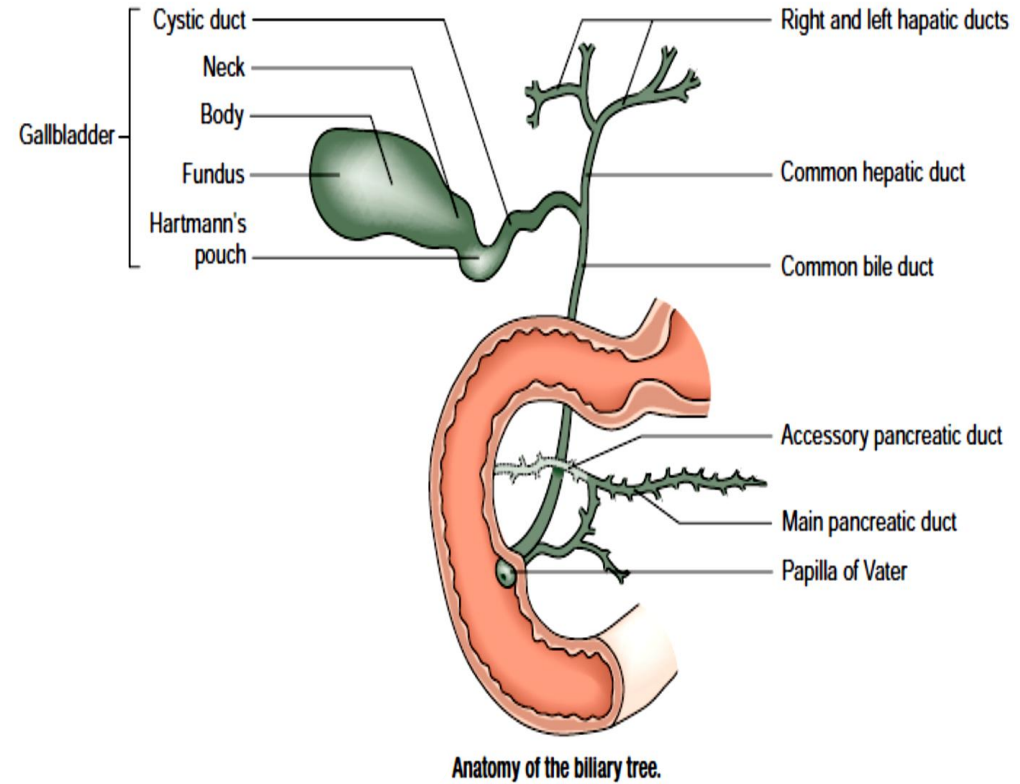
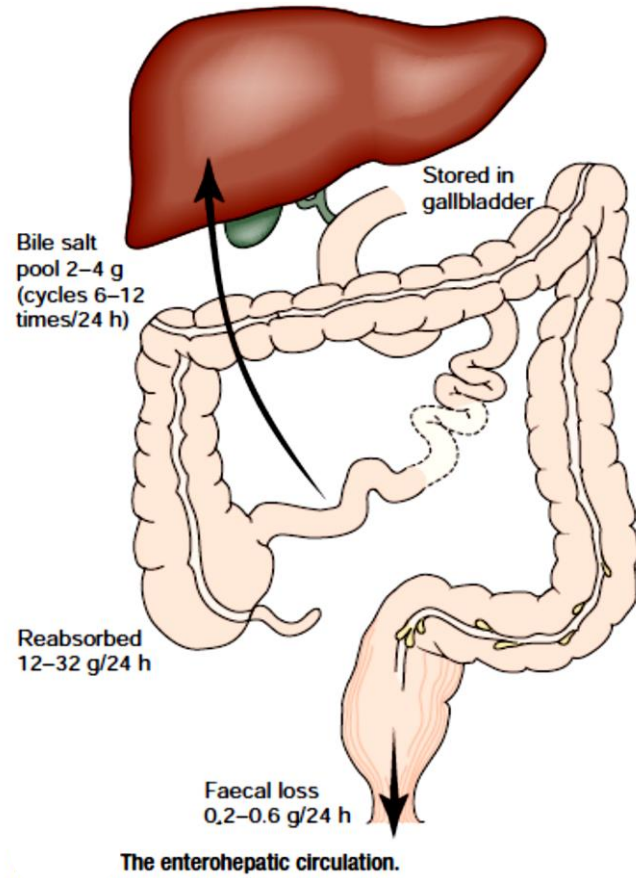




Cholecystectomy & Thyroidectomy

Dr/ Montaser Elsayy

Anatomy



Cholecystectomy

Simple cholecystectomy (removal of gall bladder and proximal cystic duct) is restricted to benign disease.

Open approach (elective) is usually only indicated for common bile duct exploration where laparoscopic exploration is **not** possible.

Indications (typical):

-
1. Symptomatic proven gallstones.
-
2. Symptomatic congenital abnormalities of the gall bladder.
-
3. Previous acalculous cholecystitis.
-

Pre-theatre preparation



- Blood Group and save required.



- Ensure **consent** is obtained for proceeding to other procedures if they are anticipated (e.g. common bile duct exploration).



- Check LFTs and any previous ERCP/MRCP imaging has been reviewed.

Positioning and theatre set-up



- Urethral catheterization.



- **Table positioning: Supine.**

Some surgeons prefer to stand between the legs for the dissection (requires the patient in leg extensions).



- Monitor/stack position. LUQ.



Patient may be placed slightly head up and left side down to improve RUQ exposure

Steps of surgery

- **Establish laparoscopy if appropriate.**
- ***Incision.*** Periumbilical; usually curved infra-umbilical although supraumbilical is also used.
- Vertical infra-umbilical can be used, especially where conversion to a midline laparotomy is anticipated.
- ***Exposure of the linea alba.*** By sharp dissection.
- ***Incision of linea alba.*** Elevate with forceps and incision with scalpel (no. 11 or 15).

Steps of surgery

Open Trochar Insertion:

- Elevate linea alba with forceps, blunt scissor opening of pre-umbilical fat pad and peritoneum and placement of trochar (blunt) or:

Blunt Trochar Insertion: Elevate linea alba with forceps without a small initial incision, insert trochar (blunt or with visual assistance using laparoscope inside the port) or:

Verres Needle Insertion:

Elevate linea alba with forceps, insert Verres needle using only thumb and finger pressure until 'clink' felt, test for intraperitoneal placement with saline 'drop' test.



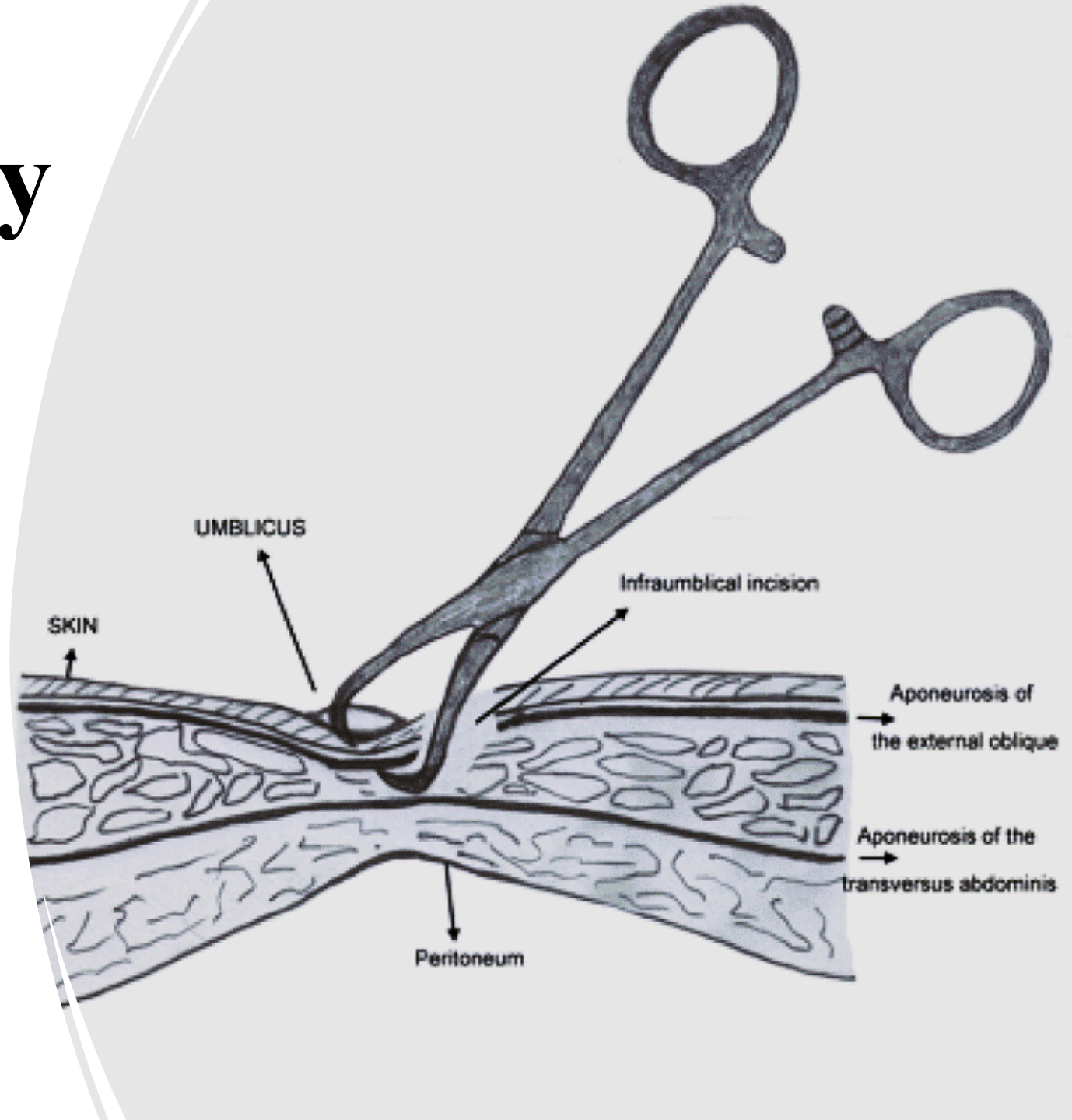
Steps of Surgery

Insufflation:

- CO₂ typical pressure between 12–15mmHg
- Use slow flow initially, check for low pressure flow before increasing flow rate.

Assessment:

- Inspect area beneath insertion port for signs of visceral injury or bleeding
- Assess anterior abdominal wall for availability of further port sites, inspect viscera sequentially.

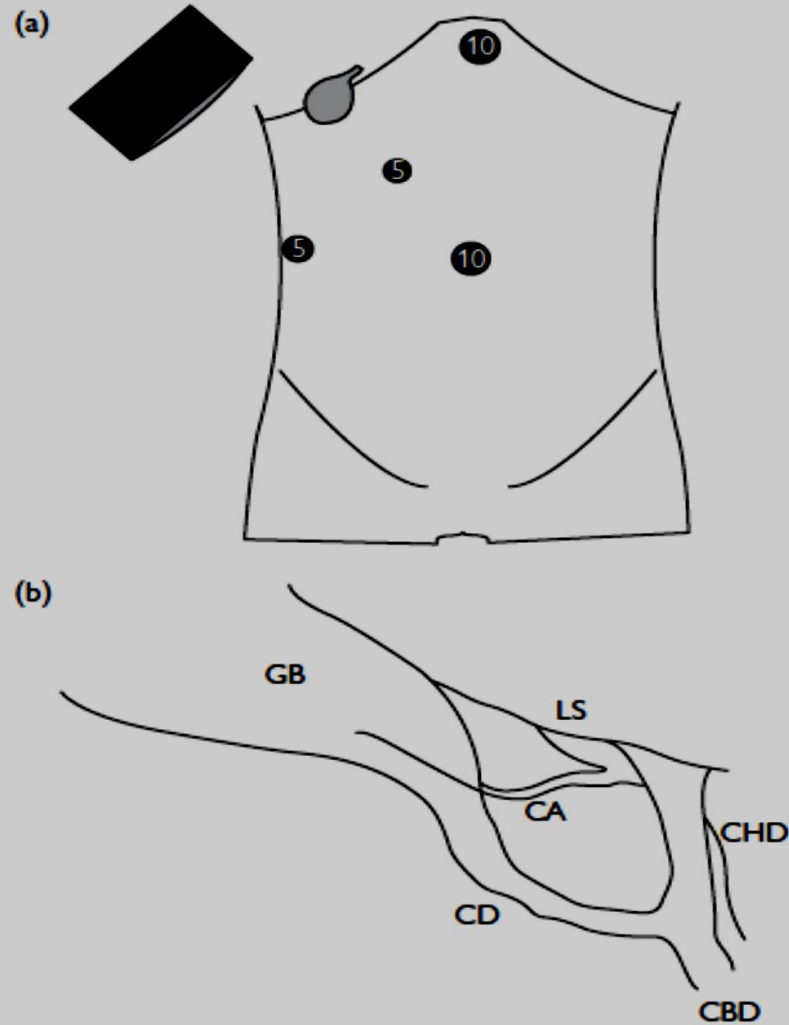


Steps

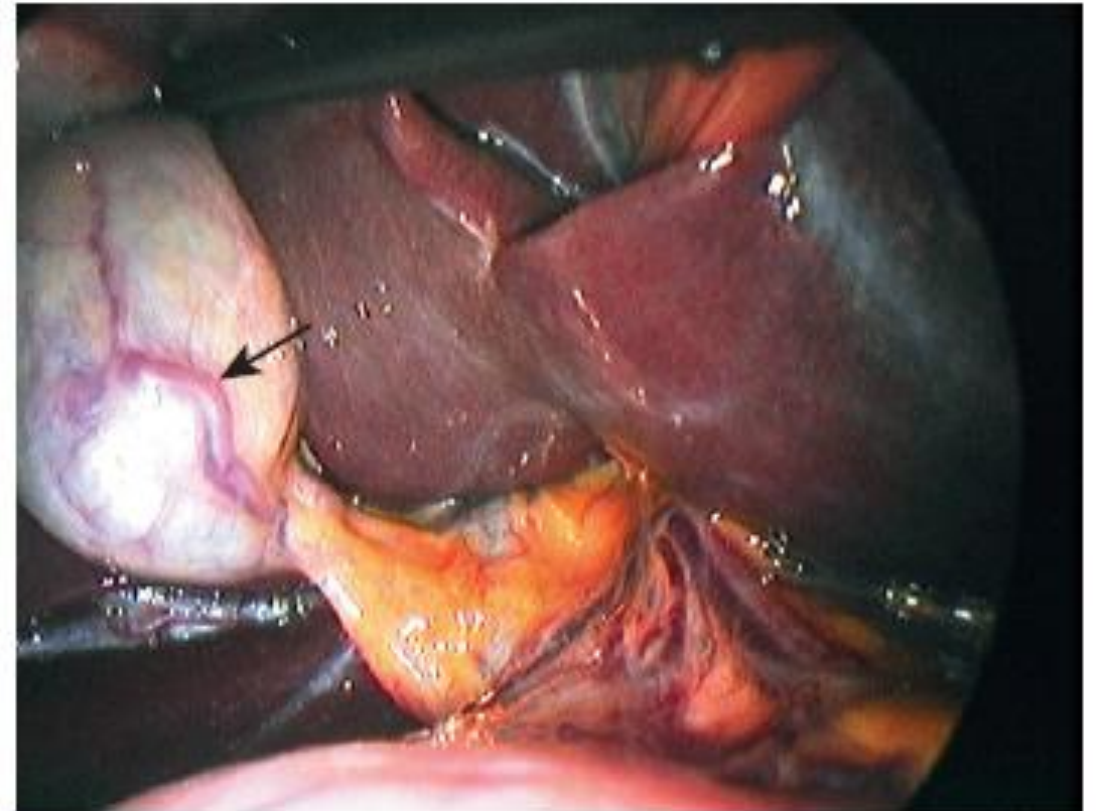
- **Incision** for open surgery: Right subcostal ('Kocher's') with (partial) division of upper right rectus muscle.
- **Expose** gall bladder neck, cystic duct and common bile duct ('Calot's triangle')—fundal traction.
- **Identify** cystic duct origin from gall bladder neck by blunt dissection.
- **Identify** cystic artery by blunt dissection.
- **Clip** (or tie if open) cystic duct (2 distal, 1 or 2 proximal on gall bladder neck) and divided.
- Repeat for cystic artery.

Steps

- **Dissect** gall bladder from liver 'bed' by retrograde dissection (i.e. from neck to fundus).
- Dissection is carried out close to the gall bladder wall.
- Gall bladder retrieved (usually in a water proof bag if laparoscopic via epigastric port).
- Check haemostasis.
- **Closure**
- Port sites: Port sites 10 mm and above require musculofascial closure, 5 mm ports do not.
- **Open:** Usually closed in layers (peritoneum, (musculo)fascial, and skin).



(a) Ports sites for laparoscopic cholecystectomy. (b) View of Calot's triangle after dissection prior to structure division. (Borders: LS, liver surface; CHD, common hepatic duct; CD cystic duct. Other structures: GB, gall bladder; CBD, common bile duct; CA, cystic artery.)



Laparoscopic view of the gallbladder attached to the main bile duct beneath the liver and showing the branches of the cystic artery (arrow) on the gallbladder surface.

Post-operative Care and Instructions

- **Remove catheter** unless required for post-operative fluid balance observation.
- **Antibiotics.** Only required for pathology found (e.g. cholecystitis).
- **Oral diet.** Normal as soon as tolerated.

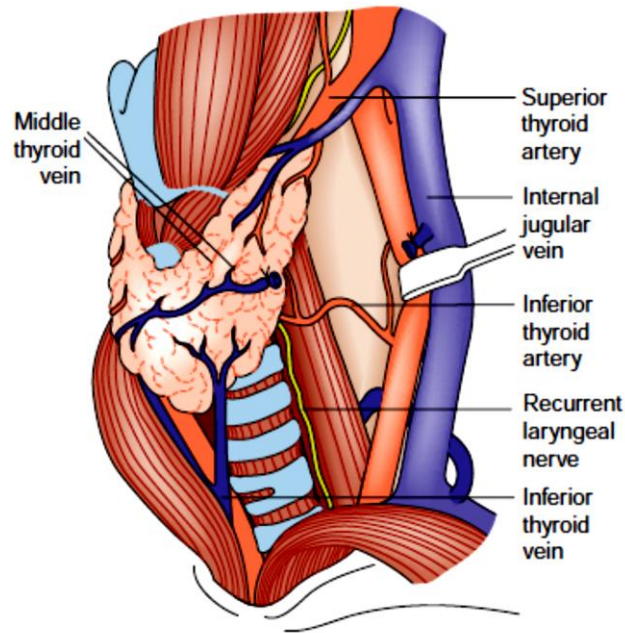
Complications (specific to the procedure)

- Port site infection, <5%.
- Bleeding, 2%.
- Conversion to open surgery (if laparoscopic), 2% (depends on gender, age, sex, previous inflammatory episodes, previous abdominal surgery).
- Visceral injury during port insertion/basic laparoscopy and assessment, 1%.
- Common bile duct injury

Cholecystectomy

- Cholecystectomy is the standard treatment for symptomatic gallbladder stones; alternatives (stone dissolution, extracorporeal lithotripsy) are now seldom used
- Open cholecystectomy has been largely superseded by laparoscopic cholecystectomy, but conversion to open operation is still sometimes needed
- Cholecystectomy now has a low operative mortality (0.2%); inadvertent injury to the bile duct (0.2% incidence) remains the main source of major morbidity
- Some 5–10% of patients undergoing cholecystectomy have ductal stones, many of which are unsuspected. Opinions vary as to whether intraoperative cholangiography should be undertaken routinely to detect such stones
- In the era of laparoscopic cholecystectomy, there is a growing tendency not to perform routine operative cholangiography, and to extract symptomatic duct stones by non-operative means (i.e. at endoscopic papillotomy)
- If ductal stones cause symptoms, they frequently give rise to cholangitis and Charcot's triad of pain, jaundice and fever (often with rigors).

Thyroidectomy



Anatomy of the thyroid gland. The middle thyroid vein has been divided to allow forward rotation of the left lobe of the gland.

I-Indications

بالترتيب

- 1- Toxic multinodular goitre.
- 2- Simple multinodular goitre in:
 - Large goitre with pressure manifestations.
 - Cosmetic disfigurement.
 - Malignant changes.
- 3- Toxic solitary nodule.
- 4- Retrosternal goitre.
- 5- Malignant goitre (Total or near total)
- 6- Graves':
 - After failure of medical Treatment.
 - Large goitre with pressure manifestations.

Investigations & Preoperative Preparation

- If toxic: → medical preparation (must be euthyroid).
- If exophthalmos: exophthalmos static measures for 6 months.
- Thyroid antibody titre, in thyroiditis.
- Blood for Thyrotoxicosis.
- Chest X-Ray for retrosternal extension.
- Valium at night of operation.
- Indirect laryngoscopy (detects silent RLN palsy)
→ for medico-legal purpose.
- Morphine & hyoscine.
(Instead of atropine in toxic cases as atropine causes tachycardia).



Anaesthesia: General Anaesthesia (Nitrous Oxide and Oxygen)

Steps:

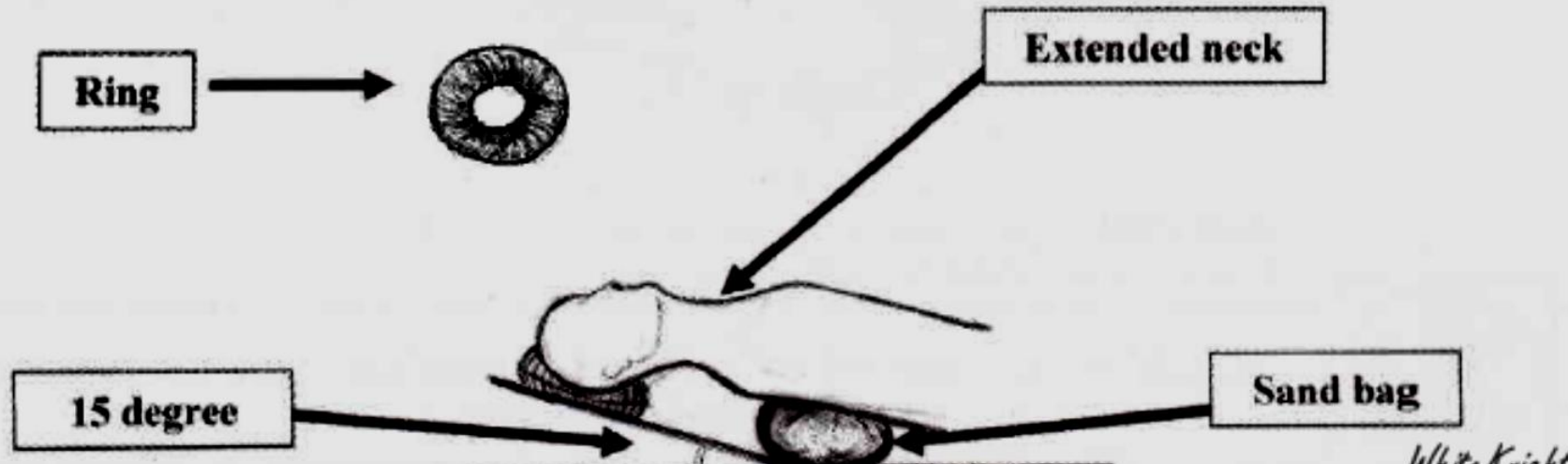
-Position of the patient

- ☞ supine
- ☞ shoulder supported on a sand bag to extend neck
- ☞ head on a ring to prevent side displacement

Of the table: → Head tilt up 15° to ↓ venous engorgement.

Adverse effect → Increases incidence of air embolism if a large vein is injured

Of the surgeon: → on the side opposite the lobe to be operated on.



Incision:

Mark the incision first with silk ligature.

Site:

- **Collar (Kocher's) incision** = 2 cm above the suprasternal notch, it extends to the posterior border of each sternomastoid.

2 cm above
sternal notch

Post. border
of
sternomastoid



NB.

1. *In large goiter, it can be extended to external jugular vein.*
2. *If the incision is done on the sternal notch, the skin of the hyperextended neck will be displaced downwards over the manubrium which will lead to keloid formation.*

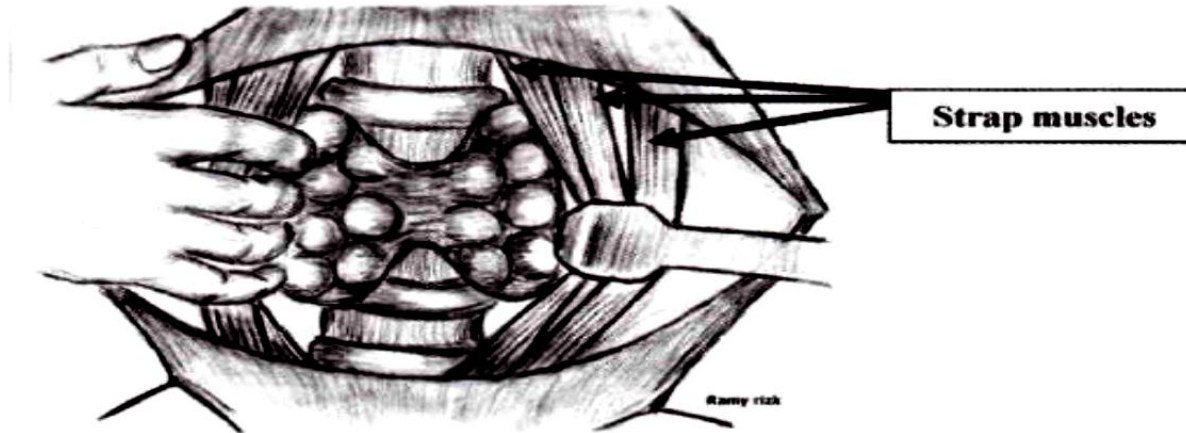
Layers:

- Incise the skin then the superficial fascia with platysma. as the blood vessels to the skin pass through platysma

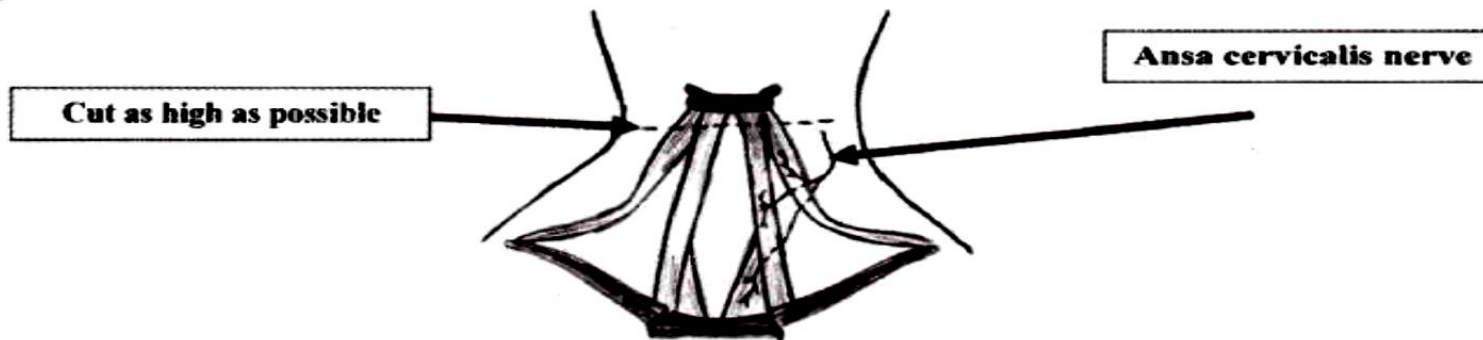


N.B. *The platysma has to be closed with the SC tissue as a separate layer to avoid adherent ugly scar.*

**Separate the strap muscles
and reflect them laterally**



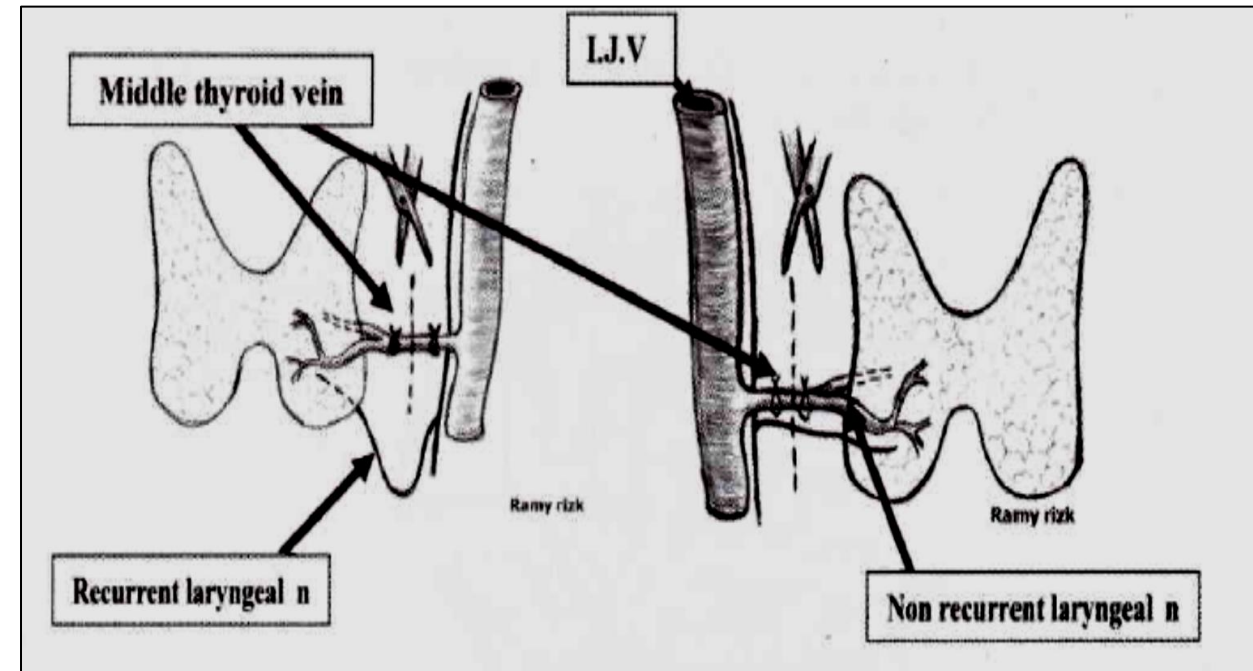
N.B. If necessary to divide the ms (in large goiter) → divide them as high as possible to preserve their nerve supply (ansa Cervicalis) which enters the lower part of the muscles.



Ligate & divide:

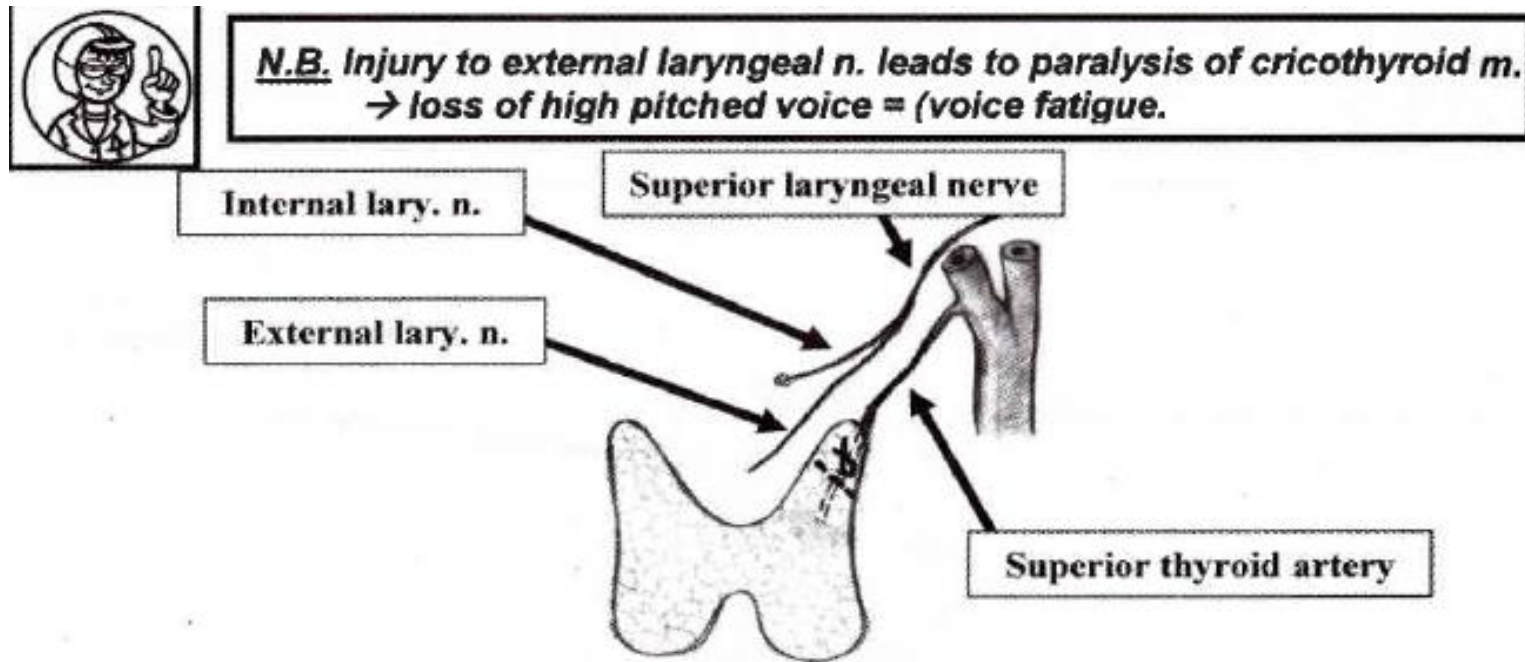
1) The middle thyroid vein:

- It is the shortest so it is the 1st to be ligated.
- In 2% of cases Rt. recurrent laryngeal n" may be non recurrent & therefore it may be injured during ligation of this vein.
- Nearer to the lower part of the gland.



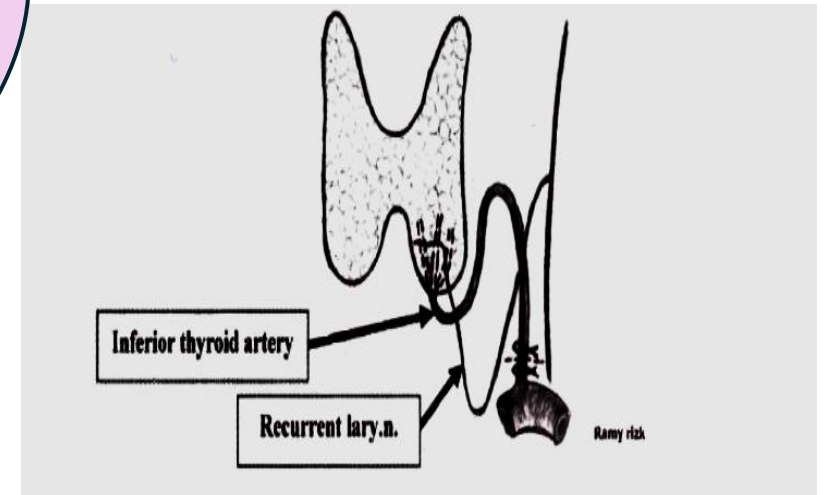
2) Superior thyroid artery & vein:

- At the **upper pole** of the gland
- **As close** as possible to the gland {& from medial to lateral) to avoid injury of **External laryngeal nerve**.



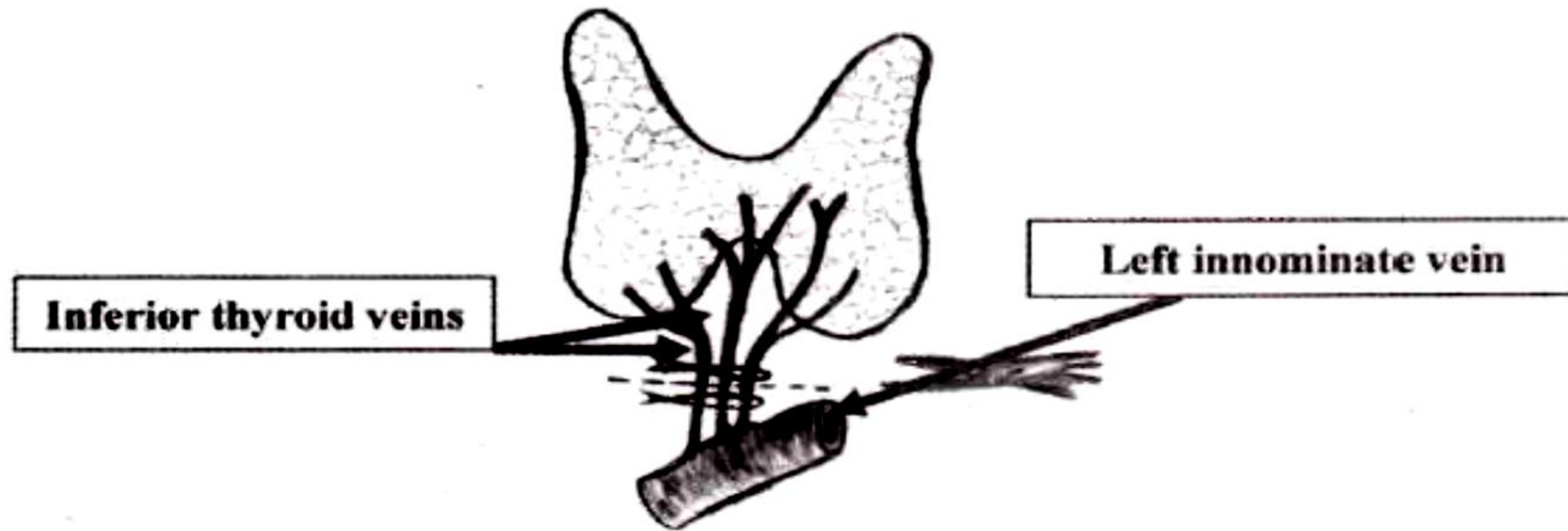
3) Inferior thyroid artery (in toxic goitre)

- At the lower pole of the gland.
- Away from** the gland (behind carotid sheath),
- To avoid injury of recurrent laryngeal n,
- It is ligated in continuity to avoid fall of proximal stump in chest (necessitates thoracotomy if slipped ligature occurs).



N.B. Some surgeons prefer to expose RLN within branches of inferior thyroid artery (i.e. near the gland) to avoid injury of blood supply of parathyroid gland.

4) **Inferior thyroid veins (3-5):** → at lower pole



N.B. Artery is ligated before vein to avoid lobe congestion.

- Delivery of thyroid lobe:

By blunt (finger) dissection from deep surface of pretracheal fascia & strap muscles,

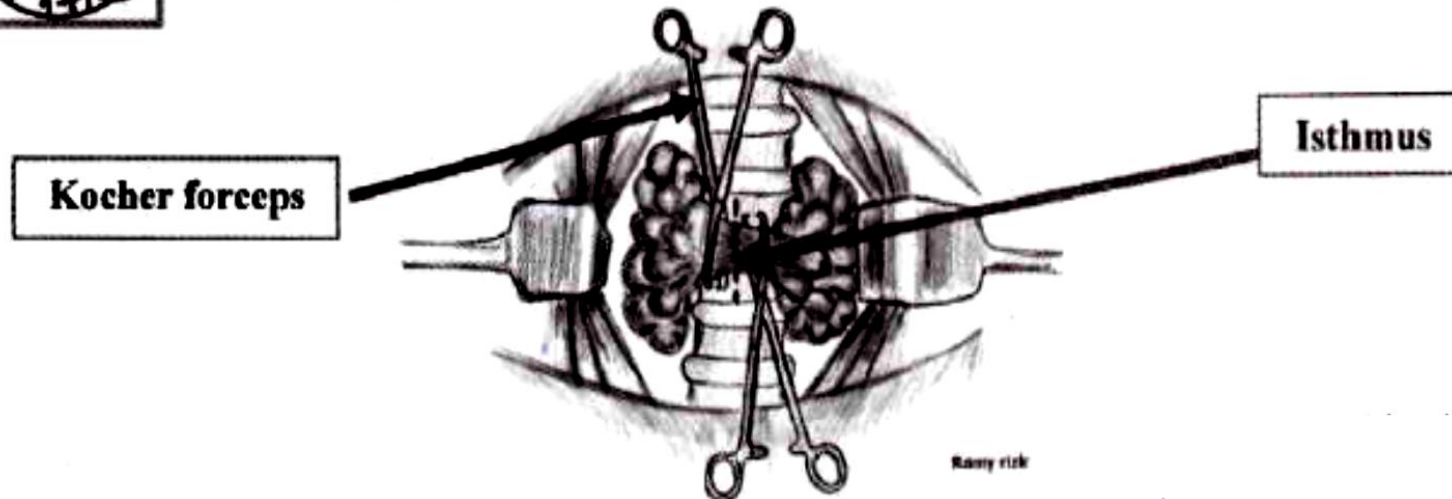
-Exposure & Devascularization of other lobe.

-Divide the isthmus Between 2 Kocher's forceps:

(This causes barring of the trachea & relief of the pressure)



N.B. Retrosternal extension is delivered by finger (sternotomy is rarely done).

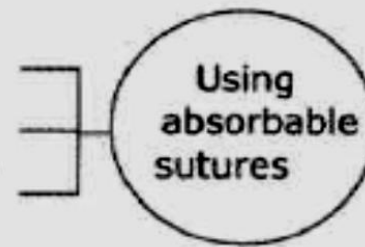


Closure

- Continuous suture to avoid haemorrhage from accessory tracheal & esophageal branches entering at ligament of Berry.
- Absolute hemostasis.
- Subfascial + SC suction drain.

- Close:

- Pretracheal fascia at the midline.
- Pretracheal muscles (if cut).
- Subcutaneous tissue including the platysma.
- Skin using:
 - a. Interrupted silk.
 - b. Subcuticular.



Postoperative Care:

- Patient in semi-sitting position.
- Care of respiration.
- Remove drains after 24 hrs.

Complications



1-**Reactionary haemorrhage:**



Cause: Due to slipped ligature from thyroid vessels, stump, flap vessels, etc.



Result: With risk of suffocation due to mechanical compression on trachea & compression of venous drainage causing laryngeal edema



Management:

Open the wound at bed side if necessary). Return the pt to operating room for haemostasis.

Complications:

2-Injury to important structures

- Trachea: → sutured by prolene (non-absorbable) as cartilage doesn't heal.
- Parathyroid: → Hypoparathyroidism.
 - Permanent:
 - Removal of the glands.
 - Temporary: from.
 - The stitches.
 - Use of cautery.
 - Bilateral ligation of inferior thyroid artery.

Injury to important structures

- Nerves:

- External Laryngeal Nerve: Cricothyroid muscle paralysis → Loss of high-pitched sound.
- Internal Laryngeal Nerve: Chocking.
- Recurrent Laryngeal Nerve: Hoarseness of voice

- Thyroid: → Hyperthyroidism or Hypothyroidism (according to the amount of the removed part).

- Internal Jugular Vein: → Air embolism.

3-Tetany

-In first 3 days..... bone hunger (corrected toxicity)

-After 1 week: hypoparathyroidism.

- C/P

Latent tetany:

- Circumoral numbness (Earliest).
- Trousseau's sign.
- Chvostek's sign.

Manifest tetany: Carpopedal spasm

Treatment:

- **Acute attack:** Calcium chloride I.V' (10 ml).
- **Later:** Vit.D (1-Alpha) + Calcium supplements

4- Thyroid Crisis: If poor preparation

5- Respiratory Obstruction:

- *Haemorrhage*
- *Laryngeal edema*
- *Bilateral vocal cord injury*
- *Tracheal injury*
- *Intraoperative aspiration*

6- Wound infection

Thank You